TYPE - A [Marks: 5]

Question - 01

1. Spanning (2.5 Marks)
States that every vector in V can be written as a linear combination of $\alpha_1, \ldots, \alpha_n$. Concludes that the set B spans V.

2. Linear Independence (2.5 Marks)
Assumes a linear combination of the vectors equals the zero vector. Uses the uniqueness of representation to conclude that all coefficients must be zero. Therefore the vectors are linearly independent.

Question - 02

1. Basis Extension (1.5 Marks)
States that the linearly independent set $\{\alpha_1, \ldots, \alpha_k\}$ with $k<n$ can be extended to a basis of V.

2. Construction (1.5 Marks)
Defines the linear transformation so that $T(\alpha_j)=\beta_j$ for $j=1,\ldots,k$. Assigns arbitrary images to the remaining basis vectors. Concludes that such a linear transformation exists.

3. Non-Uniqueness (2 Marks)
Explains that different choices for the remaining basis vectors produce different linear transformations. Therefore the transformation is not unique.
Deduct at most 0.5 marks if the student gives the general argument but does not provide an explicit counterexample.

Question - 03

(i) Showing that the set spans P_2 — 2 mark [1 mark for each step]

Step 1: Writing a general polynomial in P_2:

$f(x) = ax^2 + bx + c$

Step 2: Expressing f(x) as a linear combination of the given polynomials:

$f(x) = c_1x + c_2(x + x^2) + c_3(x + x^2 + 2)$

(ii) Linear Independence — 3 marks [1 mark for each step]

Step 1: Assuming a linear combination equals zero:

$c_1x + c_2(x + x^2) + c_3(x + x^2 + 2) = 0$

Step 2: Expanding and collecting coefficients:

$(c_2 + c_3)x^2 + (c_1 + c_2 + c_3)x + 2c_3 = 0$

Step 3: Equating coefficients and solving:

$c_1 = c_2 = c_3 = 0$

Stating that the vectors are linearly independent and span P_2.

Concluding that $\{x, x + x^2, x + x^2 + 2\}$ forms a basis of P_2.

Question - 04

1. **Solution Set (1 Mark)**
   Shows that if $T\gamma_i = \beta$ and $T\gamma_j = \beta$, then
   $$T(\gamma_i - \gamma_j) = 0 \Rightarrow \gamma_i - \gamma_j \in K$$
   where $K = \text{Null}(T)$. Concludes that all solutions are of the form $\gamma + K$.

2. **Upper Bound (1 Mark)**
   Let $k = \dim(K)$. Argues that no more than $k+1$ such vectors can be linearly independent, since otherwise their differences would produce more than k independent vectors in K, contradicting $\dim(K) = k$.

3. **Construction (1 Mark)**
   Let $\{u_1, \dots, u_k\}$ be a basis of K. Constructs the set
   $$\{\gamma, \gamma + u_1, \dots, \gamma + u_k\}.$$

4. **Proof of Independence (1.5 Marks)**
   Assumes
   $$a_0\gamma + \sum_{i=1}^{k} a_i(\gamma + u_i) = 0.$$
   Applying T gives
   $$(a_0 + \sum_{i=1}^{k} a_i)\beta = 0.$$
   Since $\beta \neq 0$, it follows that
   $$a_0 + \sum_{i=1}^{k} a_i = 0.$$
   Substituting back yields
   $$\sum_{i=1}^{k} a_i u_i = 0.$$
   Since $\{u_1, \dots, u_k\}$ are linearly independent, all coefficients are zero.

5. **Final Answer (0.5 Marks)**
   States that the maximum number of linearly independent vectors satisfying $T\gamma_i = \beta$ is
   $$\dim(\text{Null}(T)) + 1.$$

Question - 05

Consider the space V of all infinite tuples of real numbers, *i.e.* let

$$V := \{(x_1, x_2, ...) : x_1 \in \mathbb{R}, x_2 \in \mathbb{R}, ...\}$$

Let T be a "left shift operator", *i.e.* T is a linear transformation that acts as follows:

$$T(x_1, x_2, ...) := (x_2, x_3, ...)$$

i. **[2 pts] Is** $\dim(\textbf{nullspace}(T)) = 0$**? Why or why not?**

Ans. For the nullspace(T), we have to solve

$$T(x_1, x_2, ...) = (x_2, x_3, ...) = (0, 0, ...)$$

The solution is $x_2 = 0, x_3 = 0, ...$ whereas the $x_1 = t \in \mathbb{R}$. Hence, every vector in the nullspace(T) has the form $(x_1, 0, 0, ...) = x_1(1, 0, 0, ...)$. Therefore,

$$\text{nullspace}(T) = \text{span}\{(1, 0, 0, ...)\}$$

Hence, $\dim(\text{nullspace}(T)) = 1$ *i.e.* $\dim(\text{nullspace}(T)) \neq 0$.

(i) Kernel / Nullspace — 2 marks [0.5 marks for each step]

Step 1: Definition of kernel: $T(x) = 0$

Step 2: Solving the condition: $x_2 = x_3 = \cdots = 0$

Step 3: Writing the general form of vectors in the kernel: $(x_1, 0, 0, \ldots)$

Step 4: Concluding that: $\dim(\ker T) = 1$

Note: No marks deducted for writing the answer, e.g. span(1,0,0,…) with some justification.

ii. **[2 pts] Is** $\text{range}(T) \subsetneq V$**? Why or why not?**

Ans. Let $(y_1, y_2, ...) \in V$. Define $(x_1, x_2, ...) = (0, y_1, y_2, ...)$. Then

$$T(x_1, x_2, x_3, ...) = (x_2, x_3, ...) = (y_1, y_2, y_3, ...).$$

Thus every element of V has a preimage under T, and hence

$$\text{range}(T) = V.$$

Therefore, $\text{range}(T) \not\subsetneq V$.

(ii) Range — 2 marks [0.5 marks for each step]

Step 1: Writing the transformation: $T(x) = (x_2, x_3, \ldots)$

Step 2: Explaining that the output is an infinite real sequence

Step 3: Concluding that: range(T) $\subseteq$ V

Step 4: Showing that range(T) = V using the example: $x = (0, y_1, y_2, \dots)$

iii. **[1 pt] In a line, explain how your answers so far are related to the rank-nullity theorem and whether there is any contradiction here, or not.**

Ans. The rank-nullity theorem says, *Let V and W be vector spaces over the field F and let T be a linear transformation from V into W. Suppose that V is finite dimensional. Then*

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

It is given that $V := \{(x_1, x_2, ...) : x_1 \in \mathbb{R}, x_2 \in \mathbb{R}, ...\}$ *i.e.* V is infinite-dimensional and hence $\dim(V) = \infty$. Therefore, the rank-nullity theorem holds and there is no contradiction.

(iii) Rank–Nullity relation — 1 mark [0.5 marks for each step]

Step 1: Stating the Rank–Nullity Theorem

Step 2: Explaining that the theorem requires a finite-dimensional vector space.

Note: Full marks for writing $\infty + 1 = \infty$, and so the spirit of rank nullity is ok OR saying that nullity + rank is exceeding the dim(V), but rank nullity doesn't apply in infinite dimensions

TYPE - B [Marks: 8]

Question - 06

i)

- **[2 marks]** Determining Dependent Variables in E_j: $(e_{k_i} = -c_{ij})$
- **[1 mark]** Determining Free Variables in E_j: $(e_j = 1,\ e_m = 0$ for all $m \neq j)$

ii)

- **[1 mark]** Vector Space Proof of S
- **[1 mark]** Proof of Linear Independence of E_j
- **[1 mark]** Proof that E_j spans S

iii)

- **[1 mark]** Correct Dimension $(n - r)$
- **[1 mark]** Justifying the dimension (dimension of basis E_j)

Question - 07

(i) Linear Independence (3 pts)

- **1 pt: Correct setup.** Assumes

$$\sum_{i=1}^{k} x_i\alpha_i + \sum_{j=1}^{m} y_j\beta_j + \sum_{r=1}^{n} z_r\gamma_r = 0$$

  and states the goal of proving all coefficients are zero.

- **1 pt: Intersection argument.** Rearranges to obtain

$$-\sum_{r=1}^{n} z_r\gamma_r = \sum_{i=1}^{k} x_i\alpha_i + \sum_{j=1}^{m} y_j\beta_j$$

  and concludes that the left-hand side lies in both W_1 and W_2, hence in

$$W_1 \cap W_2.$$

- **1 pt: Use of basis independence.** Since $W_1 \cap W_2 = \text{span}\{\alpha_1, \ldots, \alpha_k\}$,

$$\sum_{r=1}^{n} z_r\gamma_r = \sum_{i=1}^{k} c_i\alpha_i.$$

  Using independence of the basis of W_2, conclude $z_r = 0$. Substituting back gives

$$\sum_{i=1}^{k} x_i\alpha_i + \sum_{j=1}^{m} y_j\beta_j = 0,$$

  and independence of the basis of W_1 implies $x_i = y_j = 0$.

Part - ii: 1.5 pts each.

(ii) Spanning and Dimension Identity (3 pts)

- **2 pts: Spanning argument.**
  Let $v \in W_1 + W_2$. Then
  $$v = v_1 + v_2$$
  with $v_1 \in W_1$ and $v_2 \in W_2$.
  Since
  $$W_1 = \text{span}\{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_m\}$$
  and
  $$W_2 = \text{span}\{\alpha_1, \dots, \alpha_k, \gamma_1, \dots, \gamma_n\},$$
  For any $v_1 = \sum x_i\alpha_i + \sum y_i\beta_i$, and $v_2 = \sum w_i\alpha_i + \sum z_i\gamma_i$. Then,
  $$v = v_1 + v_2 = \sum(x_i + w_i)\alpha_i + \sum y_i\beta_i + \sum z_i\gamma_i = \sum u_i\alpha_i + \sum y_i\beta_i + \sum z_i\gamma_i$$
  it follows that
  $$v \in LC(\{\alpha_i\}_i, \{\beta_i\}_i, \{\gamma_i\}_i) := SPAN(\{\alpha_i\}_i, \{\beta_i\}_i, \{\gamma_i\}_i)$$
  .

- **1 pts: Dimension identity.**
  From the construction,
  $$\dim(W_1) = k + m, \qquad \dim(W_2) = k + n, \qquad \dim(W_1 \cap W_2) = k.$$
  Since the vectors in part (i) form a basis of $W_1 + W_2$,
  $$\dim(W_1 + W_2) = k + m + n.$$
  Substituting,
  $$(k + m + n) + k - (k + m) - (k + n) = 0.$$
  Thus
  $$\dim(W_1 + W_2) + \dim(W_1 \cap W_2) - \dim(W_1) - \dim(W_2) = 0.$$
  One valid choice is
  $$c_1 = 1, \quad c_2 = 1, \quad c_3 = -1, \quad c_4 = -1.$$

(iii) Dimension Calculations (2 pts)
Given $\{\zeta_1, \zeta_2, \zeta_3, \zeta_4\}$ is a basis of V.

- **0.5 pt: Dimension of V_1.** These vectors are independent, so $\dim(V_1) = 3$.
- **0.5 pt: Dimension of V_2.** prove $d_1\zeta_1 + d_2\zeta_2$ and $d_2\zeta_1 - d_1\zeta_2$ are independent. $dim(V_3) = 3$.
- **0.5 pt: Intersection.** prove $V_1 \cap V_2 = SPAN(\zeta 1, \zeta_3)$ by proving $V_2 = SPAN(\zeta_1, \zeta_2, \zeta_3)$ $dim(V_1 \cap V_2) = 2$
- **0.5 pt: Sum dimension.** use equality from part (ii). $dim(V_1 + V_2) = 4$.

Question - 08

Part (i)

Based on Figure 1 and the problem description:

- span(A) = nullspace(T)
  The vectors in set $A(\alpha_1, ..., \alpha_k)$ all map to 0. Their span forms the nullspace of T.

- span$(B) = \text{Im}(T')$
  The problem defines $T'(\alpha_i) = \alpha_i$ for $i > k$ (which are the vectors in set B) and 0 otherwise. Therefore, the image of T' is exactly the span of the vectors in set B.

- span$(C) = \text{Im}(T)$
  Set C consists of the mapped vectors $T(\alpha_{k+1}), ..., T(\alpha_n)$. Since the first k basis vectors map to 0, the image of the entire vector space under T is spanned by the images of the remaining basis vectors.

Part (ii)

Yes, they are linearly independent.

Proof

To check for linear independence, we set up the linear combination equal to the zero vector:

$$c_1 T(\beta_1) + \cdots + c_m T(\beta_m) = 0$$

By the linearity of transformation T, we can combine the terms:

$$T(c_1\beta_1 + \cdots + c_m\beta_m) = 0$$

Let $v = c_1\beta_1 + \cdots + c_m\beta_m$. Because $T(v) = 0$, we know that $v \in$ nullspace(T). By definition, v is also a linear combination of the vectors, meaning $v \in \text{span}(\beta_1, \ldots, \beta_m)$.

Therefore, $v \in \text{span}(\beta_1, \ldots, \beta_m) \cap \text{nullspace}(T)$. Given that this intersection is strictly $\{0\}$, it follows that $v = 0$.

This gives $c_1\beta_1 + \cdots + c_m\beta_m = 0$. Since the set $\{\beta_1, \ldots, \beta_m\}$ is linearly independent, the only solution to this equation is $c_1 = c_2 = \cdots = c_m = 0$. Because all coefficients must be zero, the set $\{T(\beta_1), \ldots, T(\beta_m)\}$ is linearly independent.

Part (iii)

The statement is **True**.

Proof

From the problem description, V has a finite basis of n elements, so $\dim(V) = n$. By the Rank-Nullity Theorem:

$$\dim(V) = \dim(\text{nullspace}(T)) + \dim(\text{range}(T))$$

Given that $\text{nullspace}(T) = \{0\}$, it follows that $\dim(\text{nullspace}(T)) = 0$. Substituting this gives $\dim(\text{range}(T)) = \dim(V)$.

Since $V = W$, then $\dim(V) = \dim(W)$. Thus, $\dim(\text{range}(T)) = \dim(W)$. Because $\text{range}(T)$ is a subspace of W and has the same finite dimension as W, it must be that $\text{range}(T) = W$.

Part	**Criteria**	**Points**	**Deductions**
(i)	Correctly identifies $\text{span}(A) = \text{nullspace}(T)$	1	-1 pt if incorrect or left blank
	Correctly identifies $\text{span}(B) = \text{Im}(T')$	1	-1 pt if incorrect or left blank
	Correctly identifies $\text{span}(C) = \text{Im}(T)$	1	-1 pt if incorrect or left blank
(ii)	Sets up the linear combination and uses linearity: $T(\sum c_i\beta_i) = 0$	1	-1 pt if linearity is not shown
	Establishes vector is in intersection and concludes $\sum c_i\beta_i = 0$	1	-1 pt if intersection logic is skipped
	Uses linear independence of β_i to conclude $c_i = 0$	1	-0.5 pts if proof ends abruptly; -3 pts if only "Yes" written
(iii)	Invokes Rank-Nullity correctly to show $\dim(\text{range}(T)) = \dim(V)$	1	-1 pt if Rank-Nullity not used or incorrect
	Uses $V = W$ to equate dimensions and concludes subspace equals whole space	1	-0.5 pts if forgets to state subspace rule; -2 pts if only "True" written

TYPE - C [Marks: 10]

Question - 09

Let $\{v_1, v_2, \ldots, v_k\}$ be a linearly independent set in V.

Since $\{\beta_1, \ldots, \beta_m\}$ spans V, each vector v_j can be written as

$$v_j = \sum_{i=1}^{m} a_{ij}\beta_i, \qquad j = 1, \ldots, k.$$

Consider a linear combination

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0.$$

Substituting the expressions for v_j gives

$$\sum_{j=1}^{k} c_j \left(\sum_{i=1}^{m} a_{ij}\beta_i \right) = 0.$$

Rearranging,

$$\sum_{i=1}^{m} \left(\sum_{j=1}^{k} a_{ij}c_j \right) \beta_i = 0.$$

This yields a homogeneous system of m equations in k unknowns:

$$\sum_{j=1}^{k} a_{ij}c_j = 0, \qquad i = 1, \ldots, m.$$

If $k > m$, the system has more unknowns than equations and therefore has a non-trivial solution. Thus there exist $c_1, \ldots, c_k$, not all zero, such that

$$c_1 v_1 + \cdots + c_k v_k = 0.$$

This contradicts the assumption that $\{v_1, \ldots, v_k\}$ is linearly independent.
Therefore

$$k \leq m.$$

Hence any independent set in V must be finite and cannot contain more than m vectors.

Step	Expected Work	Marks
Independent set setup	Defines $\{v_1, \ldots, v_k\} \subset V$ and states it is linearly independent.	2
Use of spanning property	Uses that $\{\beta_1, \ldots, \beta_m\}$ spans V and writes $v_j = \sum_{i=1}^{m} a_{ij}\beta_i$.	3
Linear combination argument	Considers $c_1v_1+\cdots+c_kv_k = 0$ and substitutes expressions for v_j.	1
Formation of linear system / matrix argument	Derives system $\sum_{j=1}^{k} a_{ij}c_j = 0$ or an equivalent $m \times k$ matrix argument.	2
Key theorem	States that if $k > m$ (more unknowns than equations) the homogeneous system has a non-trivial solution.	1
Contradiction with independence	Explains that a non-trivial solution contradicts linear independence.	0.5
Final conclusion	Correctly concludes $k \leq m$.	0.5

Partial Credit Guide

Student Attempt	Typical Score
Correct idea but missing matrix/system explanation	∼7/10
States independent set $\leq$ spanning set but weak justification	∼5/10
Only states result without proof	∼2/10

Question - 10

1. (**3 pts**) Form matrix A with the given vectors as rows and reduce to the Row-Reduced Echelon Matrix R:

$$A = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 3 & 4 & -2 & 5 \\ 1 & 4 & 0 & 9 \end{pmatrix} \xrightarrow{\text{RREF}} R = \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & \frac{1}{4} & \frac{11}{4} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

   should mention $W = SPAN(\alpha_1, \alpha_2, \alpha_3)$ as row space of A/R.

2. (**1 pts**)Identification of Parameters
   - **Dimension (r):** 2 (number of non-zero rows).
   - **Pivot Indices (k_i):** $k_1 = 1$, $k_2 = 2$.
   - **Non-pivot Indices (j):** $j = 3$, $j = 4$.

3. (**6 pts**)Applying Explicit Description (Formula 2-25): A vector $\beta = (b_1, b_2, b_3, b_4) \in W$ if and only if:

$$b_j = \sum_{i=1}^{r} b_{k_i} R_{ij} \text{ --- stating this gets 3 pts}$$

   solving and giving following solution correctly: 3pts

   For $j = 3$:

$$b_3 = b_1(-1) + b_2\left(\frac{1}{4}\right) \implies 4b_1 - b_2 + 4b_3 = 0$$

   For $j = 4$:

$$b_4 = b_1(-2) + b_2\left(\frac{11}{4}\right) \implies 8b_1 - 11b_2 + 4b_4 = 0$$